

RESEARCH ETHICS PROTOCOL SUBMISSION

Phase II Randomized Controlled Trial of MetaGlyX-400 (Metformin-Glipizide Fixed-Dose Combination) in Pediatric Patients with Type 2 Diabetes Mellitus

Protocol Number:	IRB-PEDI-2026-0047	Submission Date:	June 01, 2026
Study Phase:	Phase II RCT	Review Type:	Full Board (Greater Than Minimal Risk)
Principal Investigator:	Dr. Ayesha Mahmood, MD, PhD	Department:	Pediatric Endocrinology
Institution:	Children's Medical Centre, Lahore	Sponsor:	NovaMed Pharmaceuticals Ltd.
Funding Source:	Industry-Sponsored (NovaMed)	IND Number:	IND-2026-NMP-1142
Estimated Duration:	24 months	Expected Enrolment:	120 participants

1. BACKGROUND AND SCIENTIFIC RATIONALE

Type 2 Diabetes Mellitus (T2DM) in the pediatric population represents a growing public health crisis across South Asia. Recent epidemiological data from the Pakistan Diabetes Registry (2024) indicates a 34% increase in paediatric T2DM diagnoses over the preceding five years, with mean age of onset declining to 11.3 years. Current first-line pharmacotherapy (metformin monotherapy) achieves adequate glycaemic control in fewer than 48% of paediatric patients at 12 months, creating a significant unmet clinical need for combination approaches.

MetaGlyX-400 is a novel fixed-dose combination of Metformin HCl (500 mg) and Glipizide (5 mg) formulated as a scored, dispersible tablet for paediatric administration. Preclinical studies demonstrate superior glycaemic control compared to either agent alone, with a favourable hepatic safety profile in juvenile animal models at doses up to 3x the proposed clinical dose. Adult Phase I data (NovaMed Study NMP-101, 2025) confirm acceptable tolerability; no adult Phase II data are yet available.

Knowledge Gap: No fixed-dose metformin-sulfonylurea combination has been studied in paediatric T2DM. This protocol addresses that gap while acknowledging that hepatic monitoring requirements in the paediatric population may differ from adult data.

2. STUDY OBJECTIVES

Primary Objective:

To evaluate the efficacy of MetaGlyX-400 versus metformin monotherapy in achieving HbA1c < 7.0% at 24 weeks in paediatric patients aged 8–16 years with T2DM inadequately controlled on metformin alone.

Secondary Objectives:

- Assess change from baseline in fasting plasma glucose (FPG) at 12 and 24 weeks.
- Evaluate safety and tolerability, including hypoglycaemic episodes and hepatic function.
- Assess patient-reported quality of life using the PedsQL Diabetes Module.
- Characterise pharmacokinetic parameters in the paediatric population (sub-study, n=20).

3. STUDY DESIGN AND METHODOLOGY

This is a Phase II, randomised, double-blind, parallel-group, placebo-controlled trial conducted at three paediatric endocrinology centres in Pakistan. Participants will be randomised 1:1 to MetaGlyX-400 or metformin monotherapy (active comparator) for 24 weeks with a 4-week safety follow-up.

Eligibility Criteria:

Inclusion:

- Age 8–16 years (inclusive) at time of screening.
- Confirmed diagnosis of T2DM per ADA criteria for at least 6 months.
- HbA1c 7.5%–10.0% on stable metformin monotherapy ($\geq 1,500$ mg/day) for ≥ 3 months.
- BMI at or above the 85th percentile for age and sex.
- Written informed consent from parent/legal guardian.

Exclusion:

- Type 1 diabetes or monogenic diabetes.
- eGFR < 60 mL/min/1.73m² at screening.
- ALT or AST > 2.5 x upper limit of normal at screening.
- Known hypersensitivity to metformin or sulfonylurea class agents.
- Current use of systemic corticosteroids or other antidiabetic agents.
- Pregnancy or lactation (females of childbearing potential).

Study Procedures:

Visit	Week	Key Procedures
Screening (V0)	-2 to 0	Eligibility review, labs (HbA1c, FPG, LFTs, eGFR), ECG
Baseline (V1)	0	Randomisation, study drug dispensing, PedsQL, PK sub-study enrol.
V2	4	Safety labs, hypoglycaemia diary review, AE assessment
V3	12	HbA1c, FPG, LFTs, PedsQL, dose adjustment if required
V4 (Primary)	24	HbA1c (primary endpoint), FPG, LFTs, full safety panel, PedsQL
Follow-up	28	Safety follow-up, SAE review, end-of-study procedures

4. RISK ASSESSMENT AND BENEFIT ANALYSIS

This study involves a VULNERABLE POPULATION (minors aged 8–16 years) receiving an investigational pharmacological intervention. Risk classification: **GREATER THAN MINIMAL RISK** per 45 CFR 46.405.

Risk Category	Specific Risk	Likelihood	Severity	Mitigation
Hypoglycaemia	Symptomatic low blood glucose from long-acting insulin glargine component	Moderate	Moderate	Glucose monitoring diary; hypoglycaemia rescue protocol; dose held if hypoglycaemia
Hepatotoxicity	Drug-induced liver injury (DILI) — glipizide class effect	Low	High	LFT monitoring at Weeks 4, 12, 24. Stopping rule: ALT/AST > 3x ULN
Hypersensitivity	Allergic reaction to study drug components	Low	Variable	Exclusion of known hypersensitivity; anaphylaxis protocol at site
Burden	Blood draws, additional clinic visits	Certain	Low	Minimised venepuncture; visit windows allow flexibility

Benefit Assessment: Participants may experience direct clinical benefit through improved glycaemic control. Indirect benefits include close medical monitoring throughout the trial period. The potential benefits are considered to justify the risks described above in the context of an inadequately controlled chronic condition in a growing child.

5. INFORMED CONSENT PROCEDURES

Given that all participants are minors, a two-stage consent process will be employed in accordance with institutional policy and applicable regulations.

5.1 Parental / Guardian Permission

Written informed permission will be obtained from a parent or legal guardian prior to any study procedures. The permission document has been developed in Urdu and English and covers: study purpose, procedures, duration, foreseeable risks (hypoglycaemia, blood draws), confidentiality, voluntary participation, and right to withdraw. A study physician will review the document with the family and answer all questions. A minimum 24-hour reflection period will be provided before signing.

5.2 Participant Assent

For younger participants (ages 8–11), verbal explanation of the study will be provided in age-appropriate language by a trained research nurse. Participants will be encouraged to ask questions and will be informed that they may stop at any time. Given the young age of participants, formal written assent is not required for this age group.

Note: The site research team will use a pictorial explanation booklet titled 'What is a Clinical Trial?' for all paediatric participants to support understanding.

5.3 Consent for Pharmacokinetic Sub-Study

Participants in the PK sub-study will undergo two additional blood draws (10 mL each) at Week 12. Separate parental permission for the sub-study will be obtained. Participation in the sub-study is optional and does not affect eligibility for the main trial.

6. DATA PRIVACY AND MANAGEMENT

6.1 Data De-identification

All participant data will be de-identified using a coded participant ID system (format: PEDI-XXXX). The master linkage file linking participant IDs to personal identifiers will be maintained separately by the Principal Investigator and accessible only to essential study personnel. De-identification meets the HIPAA Safe Harbour standard (45 CFR 164.514(b)).

6.2 Data Storage

Electronic data will be captured in a validated electronic data capture (EDC) system (MedRio EDC, Version 4.2). Data will be stored on servers located within Pakistan. Access controls, audit trails, and role-based permissions are enabled. Physical study documents will be stored in locked cabinets at each site for a minimum of 15 years post-trial completion.

6.3 Data Sharing with Sponsor

De-identified study data will be transferred to NovaMed Pharmaceuticals Ltd. and their appointed Contract Research Organisation (CRO), BioSync Research Pvt Ltd (Karachi), for statistical analysis and regulatory submission purposes. Data transfer will occur via encrypted secure file transfer protocol.

Data sharing with BioSync Research will be governed by a standard confidentiality agreement as part of the CRO services contract.

6.4 Data Retention and Destruction

At trial completion, all identifiable data will be destroyed per institutional data governance policy. De-identified data may be retained indefinitely for future research purposes, as stated in the parental permission document.

7. ADVERSE EVENT MONITORING AND REPORTING

All adverse events (AEs) will be recorded from the time of first dose until the end of the safety follow-up period (Week 28). AEs will be graded using the CTCAE v5.0 and assessed for causality, severity, and expectedness.

Serious Adverse Event (SAE) Reporting:

- SAEs will be reported to the IRB within 24 hours of the investigator becoming aware.
- Expedited safety reports will be submitted to DRAP (Drug Regulatory Authority Pakistan) within 7 calendar days for fatal or life-threatening SAEs and 15 calendar days for all other SAEs.
- NovaMed Pharmacovigilance will receive SAE reports within 24 hours per the Safety Management Plan.
- An independent Data Safety Monitoring Board (DSMB) will conduct interim safety reviews at enrolment of 40 and 80 participants.

Stopping Rules: The trial will be stopped early if the DSMB identifies a statistically significant excess of hypoglycaemic events requiring hospitalisation in the MetaGlyX-400 arm ($p < 0.001$ at interim analysis). Individual participant discontinuation criteria include: withdrawal of consent, protocol deviation at investigator discretion, or pregnancy.

8. STATISTICAL ANALYSIS PLAN

The primary analysis will compare the proportion of participants achieving HbA1c < 7.0% at Week 24 between treatment arms using a chi-squared test. The sample size of 120 participants (60 per arm) provides 80% power to detect a 20% difference in response rate (35% vs 55%) at a two-sided alpha of 0.05, assuming 15% dropout.

The primary analysis population will be the modified intent-to-treat (mITT) population, defined as all randomised participants who received at least one dose of study drug and had at least one post-baseline HbA1c measurement. Missing data will be handled using multiple imputation under a missing-at-random assumption.

Secondary endpoints will be analysed using mixed-model repeated measures (MMRM) for continuous variables and logistic regression for binary outcomes. All analyses will be performed by BioSync Research Pvt Ltd using SAS Version 9.4.

9. ETHICAL CONSIDERATIONS

This study will be conducted in accordance with the Declaration of Helsinki (2013 revision), ICH E6(R2) Good Clinical Practice guidelines, and applicable Pakistani regulatory requirements under DRAP. The study received scientific review and approval from the Children's Medical Centre Research Committee on March 14, 2026.

Special Considerations — Vulnerable Population:

- All recruitment will be through existing clinical relationships; no direct-to-public advertising targeting minors.
- Financial compensation to families will not exceed reimbursement of documented transport and meal costs to avoid undue inducement.
- A participant advocate (independent of the research team) will be available at each site visit.
- Results will be communicated to participants and families in lay language at trial completion.

10. INVESTIGATOR DECLARATIONS

I, the undersigned Principal Investigator, confirm that the information provided in this protocol is accurate and complete to the best of my knowledge. I agree to conduct this study in accordance with the approved protocol, applicable regulations, and ICH GCP.

Principal Investigator	Co-Investigator	Department Head
Dr. Ayesha Mahmood, MD, PhD	Dr. Usman Tariq, MD	Prof. Nadia Hussain, FRCP
Paediatric Endocrinology	Clinical Pharmacology	Department of Paediatrics
Date: June 01, 2026	Date: _____	Date: _____

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Date Received: _____ Assigned Reviewer(s): _____

Review Outcome: ☐ Approved ☐ Approved with Modifications ☐ Deferred ☐ Disapproved

IRB Chair Signature: _____ Date: _____

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